

Abdullah Al Noman

Phone: +1-682-716-2745 Email: abdullahal.noman@uta.edu LinkedIn: [linkedin.com/in/noman-abdullah-al](https://www.linkedin.com/in/noman-abdullah-al)
Website: Abdullah Al Noman

Education

Ph.D. in Computer Science <i>University of Texas at Arlington, USA</i> <i>Transformative Wireless Systems and Technology (TWiST) Lab</i> PhD Supervisor: Dr. Debashri Roy	2025 – Present CGPA: 3.71/4.0
Research Focus: Space–Air–Ground Integrated Networks (SAGIN), Direct-to-Satellite (D2S), Non-Terrestrial Networks (NTN), O-RAN (Open Radio Access Network), Next-Generation (NextG) Wireless Networks	
Bachelor of Technology (B.Tech.) in Computer Science & Engineering <i>National Institute of Technology Tiruchirappalli, India</i>	2020 – 2024 CGPA: 7.51 / 10.00 (75.1%)

Publications

- M. Kulhandjian, **A. A. Noman**, and D. Roy. "Graph-Spectral Identifiability and Cramér–Rao Bounds for Multistatic TDOA Geolocation with Uncertain Sensor Poses," *IEEE Military Communications Conference (MILCOM)*, 2026, **Accepted**.
- M. Kulhandjian, **A. A. Noman**, H. Kulhandjian, and D. Roy. "Direct Position Determination of Non-Cooperative Emitters in the Delay–Doppler Domain," *60th Asilomar Conference on Signals, Systems, and Computers*, 2026, **Accepted**.
- T. T. Sari, R. Ahmed, **A. A. Noman**, G. Seçinti, C. Dick, and D. Roy. "TITAN: Twin-Informed Topology Adaptation for LAWN-enabled D2C Communication," *arXiv preprint*, 2026.
DOI: arXiv preprint
- **A. A. Noman**, T. T. Sari, S. Amatare, G. Seçinti, and D. Roy. "Space-Air-Ground Network for Direct-to-Cell Communication," *IEEE Military Communications Conference (MILCOM)*, 2025.
DOI: document/11310652
- M. Priyadarsini, **A. A. Noman**, V. Patel, and S. Prakash. "A Reliable and Secure Multicast Transport Protocol for Space-Ground Integrated Networks using RBAC," *15th International Conference on Computing Communication and Networking Technologies (ICCCNT), IEEE*, 2024.
DOI:10.1109/ICCCNT61001.2024.10725800

Experience

Graduate Teaching Assistant <i>University of Texas at Arlington</i>	Jan 2025 – Arlington, USA
• Supported undergraduate and graduate course instruction by assisting with lectures, laboratory sessions, technical discussions, and guided problem-solving activities. • Evaluated assignments, exams, projects, and lab reports using standardized rubrics while providing constructive feedback and maintaining accurate academic records.	
Machine Learning Intern <i>Indian Institute of Technology (IIT) Ropar</i>	May 2023 – July 2023 Punjab, India
• Developed and implemented machine learning algorithms for image enhancement and green screen removal tasks, optimizing real-time processing. • Achieved significant improvements in processing time and accuracy (93%) compared to existing models, leading to more efficient solutions.	
Summer Research Intern <i>National Institute of Technology Tiruchirappalli</i>	May 2022 – Aug 2022 Tamil Nadu, India

- Led the development of an **Intrusion Detection System (IDS)** for smart homes using Raspberry Pi, leveraging IoT devices to monitor and detect unauthorized activities.
- Collected and analyzed data from a network of sensors, integrating with a central control unit to trigger alarms and send notifications in case of intrusion.

Projects

Development of an O-RAN System | *Open Air Interface (OAI) 5G NR, Near-RT RIC, xApps*

- Actively contributing to the design and implementation of an O-RAN (Open Radio Access Network) system at TWiST lab to enhance network flexibility and interoperability.
- Developing and optimizing the RAN Intelligent Controller (RIC) for real-time decision-making in wireless communication.

Spherical Interpolation for Real-Time Emitter Navigation | *Software-Defined Radio, OctoClock, MATLAB*

- Built a synchronized 4-node USRP SDR system using OctoClock to enable coherent IQ capture and TDOA-based localization.
- Implemented and evaluated a mobility-robust Spherical Interpolation (SI) localization pipeline, validated through indoor static and mobile experiments.

Secure and Reliable Multicast Transport Protocol for Space-Ground Integrated Networks | *NS-2, Python, Secure Multicast*

- Designed a reliable and secure multicast transport protocol for space-ground integrated networks using role-based access control (RBAC) to mitigate security and reliability vulnerabilities.
- Implemented an intermediary-based communication framework with automated role verification and anomaly detection, enabling fault tolerance through backup intermediary activation.
- Evaluated system performance through experimental analysis, achieving 13.5 Mbps throughput, 0.004% packet delivery ratio (PDR), low transmission delay, and improved anomaly detection accuracy.

Technical Skills

Programming Languages: C, C++, Python, Dart

Web Technologies: HTML, CSS, JavaScript

Libraries & Frameworks: NVIDIA Sionna, TensorFlow, PyTorch, OpenCV, Matplotlib, pandas, Keras

Engineering Tools: MATLAB, Weka, Unity, Jupyter Notebook, VS Code, Android Studio, Makefile, Git

Reviewer Roles

- IEEE/ACM Transactions on Networking(TNET)
- IEEE 103rd Vehicular Technology Conference: VTC2026-Spring
- IEEE Transactions on Mobile Computing (TMC)
- Elsevier Computer Communications
- IEEE Global Communications Conference (GLOBECOM 2025)
- The 21st International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob 2025)
- The 18th International Workshop on Selected Topics in Wireless and Mobile Computing (STWiMob 2025)
- The 34th International Conference on Computer Communications and Networks (ICCCN 2025)
- EuCNC & 6G Summit (2025)

Achievements

- Awarded the prestigious Indian Council for Cultural Relations (**ICCR**) scholarship by the Government of India to pursue a B.Tech in Computer Science & Engineering in India.
- Honored with the Islami Bank Merit Scholarship for exceptional performance in the 10th standard board exam.
- Granted a government scholarship by the People's Republic of Bangladesh for achieving above 90% marks in both the 5th and 8th grades.